

Notes on Classical Electron Dynamics: Lorentz–Dirac Equation

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ABSTRACT

1. THE LORENTZ EQUATION AND THE RUNAWAY PROBLEM

[Conventions, stated once here and used throughout: Gaussian units for electrodynamics, SI for mechanics, $c = 1$. Metric signature and the definition of u^α , a^α to be fixed here. Note that the 2006 text declared Gaussian units but evaluated τ_0 without the factor $1/4\pi\epsilon_0$; the corrected value is used here from the start.]

1.1. *Electrodynamics does not supply an equation of motion for the charge*

Classical electrodynamics rests on three pillars: the conservation laws, chief among them the conservation of charge; Maxwell’s equations; and the Lorentz force law. Maxwell’s equations fix the field produced by a given distribution of charge and current. The Lorentz force law fixes the force exerted on a charge by a given field. Between them the two appear to account for everything an electrodynamic problem could ask for, and they do not: neither supplies the equation of motion of the charge itself.

The gap is structural rather than an oversight. To follow the motion of a charge one needs the force acting on it, hence the total field at its location, and the total field includes the field that the charge itself produces. Maxwell’s equations say how that self-field is generated but not how it acts back; the Lorentz force law says how a field acts on a charge but is silent on whether the charge’s own field belongs among the fields that act. Electrodynamics is complete as an account of how charges make fields and of how fields push charges, and it is incomplete exactly where the two meet on the same particle. Closing that gap is the subject of everything below.

1.2. *Three historical routes*

Any attempt to close the gap must begin by deciding what a charge is, and two pictures divide the subject between them. In the first the charge sits at a point. Its own field then diverges as the point is approached, and with the field the Lorentz force, so the force expression loses its meaning exactly where it is needed. One may object that the Lorentz force was only ever intended for external fields and that the self-field is simply not subject to it; but the objection buys nothing, because it leaves the self-force undetermined and the equation of motion no closer than before. That is the standing difficulty of the point picture. In the second the charge is spread over a finite region and treated as a continuous medium. Every quantity is then finite and the self-force is a well-defined integral over the distribution. The price is that like charges repel, so a body carrying charge of one sign must fly apart; bound charges nevertheless exist, and electrodynamics does not say what holds them together.

Three attempts, spread over thirty years, took these pictures in turn. Poincaré (1906) worked in the extended picture and supplied the missing agent by hand, adding to the electromagnetic stress–

energy tensor a binding tensor — the Poincaré stresses — chosen so that the total is in equilibrium and the charged body holds. The construction does what is asked of it, but it postulates a force for the sole purpose of removing a single contradiction, with no other consequence and no independent evidence for its existence; in the century since, few have followed it.

Lorentz (1909) took a position between the two pictures: the electron is a sphere of finite radius carrying a uniform charge. This is concrete enough to yield a definite equation of motion, which is the subject of Sec. 1.3.

Dirac (1938) returned to the point picture, but gave up computing the self-force as the mutual action of one part of the charge on another. He imposed momentum–energy conservation on a thin tube surrounding the world line and read the equation of motion off the balance, so that the divergent self-field enters only through quantities that cancel. This is the route the rest of these notes follow; it is set out in Sec. 2.

These notes adopt the point picture, for the reasons that recommended it to Dirac. It is technically uniform; the model carries no internal structure that has to be specified in advance; the equation of motion comes out in a fixed form; and the construction generalizes, both to curved backgrounds and to the corresponding gravitational problem. The cost is that infinite quantities appear at intermediate stages, so that the force has to be extracted indirectly — either by balancing finite quantities, as in the conservation method, or by separating off a singular part and discarding it. In what sense those operations are legitimate is a question that runs through everything below.

1.3. *The Lorentz model and its three defects*

Lorentz’s electron is a sphere of finite radius carrying a uniform charge: far enough into the extended picture that nothing diverges, structureless enough to remain tractable. What holds the charge together is left unanswered, which is precisely the question the Poincaré stresses were invented to settle. In motion the sphere is contracted into an ellipsoid.

Two forces then act on it. One is external. The other is a self-force, produced by the action of the charge elements on one another. At rest this self-force vanishes by symmetry. In motion it does not, and the reason is retardation: the field with which one element acts on a second is the field it carried at an earlier time, and the reaction of the second on the first is delayed in the same way, so the two no longer cancel. The charge elements taken by themselves therefore violate Newton’s third law. Nothing is genuinely lost — the missing momentum resides in the field — but the failure is a fair measure of how far the problem has already moved away from mechanics.

Expanding the retarded interaction in powers of the light-crossing time of the sphere produces, in order: a term proportional to $\dot{\mathbf{v}}$ whose coefficient grows without bound as the radius shrinks; a term proportional to $\ddot{\mathbf{v}}$ whose coefficient does not depend on the radius at all; and further terms carrying positive powers of the radius, which vanish in the point limit. Absorbing the first into the mass leaves

$$m\dot{\mathbf{v}} - \frac{2}{3}e^2\ddot{\mathbf{v}} + \cdots = \mathbf{F}, \tag{1}$$

now usually called the Abraham–Lorentz equation, in which m is the observed mass and everything from the second term on the left is the self-force. Three objections can be raised against it.

The mass is entirely electromagnetic, and comes out with the wrong coefficient. The absorbed term gives $m = 4U_0/3c^2$, where U_0 is the electrostatic self-energy, finite here because the radius is. The

factor $4/3$ is itself a symptom: a covariant accounting of the momentum of a bound charge distribution yields U_0/c^2 , and the extra third appears only because the electromagnetic stress has been counted while whatever holds the sphere together has not — the same omission as before, now showing up as a number. Fermi and Wilson later showed that the coefficient becomes unity once the binding stresses are included. The deeper objection survives the repair. If inertia is electromagnetic in origin, a neutral particle should have no mass, and neutral particles have mass. Inertial mass cannot be a manifestation of charge; at most one may say that charge brings mass with it. Nor does Lorentz's derivation claim otherwise on its own terms, since the U_0 appearing in it is electromagnetic energy alone and contains no mechanical rest energy. One way to keep the question out of the argument altogether is to obtain the damping force from a periodically moving electron, by averaging the work done against the radiated power; the mass then stays where classical mechanics put it, at the cost of yielding only an average effect, with the third and higher derivatives of $\mathbf{v}$ lost.

The equation is not covariant. Equation (1) is written with d/dt in a particular frame. The repair is due to Landau and Lifshitz and is disarmingly short: replace d/dt by $d/d\tau$ and fix the residual freedom with $u^\alpha a_\alpha = 0$. What emerges is the covariant form of the reaction term, the same one that Dirac's much longer construction produces; it is written down in Sec. 2.

A free electron accelerates itself without bound. This is the serious defect, and it is not repaired anywhere in these notes. It is the subject of Sec. 1.4.

1.4. The Abraham–Lorentz equation and its runaway solutions

[Set $\mathbf{F} = 0$ in Eq. (1), drop everything past the second term and reduce to one dimension: $ma = \frac{2}{3}(e^2/c^3)\dot{a}$, that is $\dot{a} = a/\tau_0$, with solution $a = C e^{t/\tau_0}$. Then: (i) $C \neq 0$ is a free electron accelerating itself to the speed of light with nothing acting on it; (ii) $C = 0$ restores Newton's first law, but has to be imposed by hand — the equation is first order in a and admits any $a(0)$, and demanding $a(0) = 0$ merely to keep a finite as $t \rightarrow \infty$ has no physical warrant; (iii) the impulse problem $ma - \frac{2}{3}e^2\dot{a} = \tau\delta(t)$ sharpens it: an uncharged particle struck by an impulse jumps in velocity and returns to zero acceleration, whereas here $a = 0$ before the pulse and $a = \kappa e^{t/\tau_0}$ after it. Close on the root cause: Newton's equation contains a alone, this one contains derivatives of a , so the electron goes on accelerating after the force has gone — and what keeps it accelerating is its own field.]

[Define the characteristic time here and give the number, since every later section uses it:]

$$\tau_0 = \frac{2e^2}{3mc^3} = 6.27 \times 10^{-24} \text{ s} \quad (\text{electron}). \quad (2)$$

1.5. Criteria for a physically acceptable equation of motion

[The three criteria stated in the introduction of the 2006 text, to be used as the scorecard in Secs. 3, 4 and 5: (1) the motion follows from an ordinary differential equation solved with initial/boundary data; (2) a steadily moving particle perturbed by a pulse returns to steady motion, rather than running away to c or oscillating between $\pm c$; (3) energy conservation and causality hold.]

2. THE LORENTZ–DIRAC EQUATION: DERIVATION

[Terminology: throughout these notes “Lorentz–Dirac” (LD, also called Abraham–Lorentz–Dirac) refers to the classical equation of motion of a point charge, never to Dirac's relativistic wave equation.]

2.1. The world-tube momentum–energy balance

[Dirac’s construction: surround the world line by a thin tube, integrate the conservation law over it, and let the radius shrink. Set up the geometry and the surface integrals.]

2.2. The undetermined four-vector B^μ

[The tube calculation leaves a four-vector B^μ undetermined, restricted only by $v \cdot B = 0$. Dirac’s choice is one option among many; this freedom is what Sec. 5 exploits.]

2.3. The equation

$$ma^\alpha = F_{\text{ext}}^\alpha + \frac{2}{3}e^2 (\delta^\alpha_\beta + u^\alpha u_\beta) \dot{a}^\beta. \quad (3)$$

[Identify the two pieces of the reaction term: the Schott term and the radiated-momentum term, and note that the projector enforces $u \cdot a = 0$.]

2.4. The other route: separating the singular part

[State without proof here that the same equation follows from computing the Lorentz force with $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$ as the potential. The equivalence of the two routes is proved in Sec. 4.1, and that proof is what opens the door to Sec. 4.]

2.5. Two reservations about the derivation

[As recorded in the 2006 text: (i) the form of B^μ is arbitrary, fixed only by $v \cdot B = 0$; (ii) both derivations evaluate the tube surface integral but not the cap integrals.]

3. THE LORENTZ–DIRAC EQUATION: DISCUSSION AND SOLUTIONS

3.1. One-dimensional reduction

[Parametrize by the rapidity q via $\dot{t} = \cosh q$, $\dot{x} = \sinh q$. Equation (3) collapses to a single scalar equation:]

$$\dot{q} - \frac{2}{3}\ddot{q} = F^{01}. \quad (4)$$

3.2. The free solution and the runaway

$$q = C_1 e^{3\tau/2} + C_2. \quad (5)$$

[$C_1 \neq 0$ is the runaway; $C_1 = 0$ has to be imposed by hand. Note that no initial condition on position and velocity alone selects it — the equation is third order.]

3.3. Dirac’s asymptotic condition and the price paid

[Pushing the condition out to $\tau \rightarrow \pm\infty$ eliminates the runaway but introduces pre-acceleration: the particle starts moving before the force arrives, on a timescale τ_0 . This trades criterion (2) for criterion (3) of Sec. 1.5.]

3.4. Energy balance and the Schott term; hyperbolic motion

[Where the energy sits during acceleration, and the standing puzzle of uniformly accelerated motion, in which the reaction term vanishes although the particle radiates.]

3.5. Scorecard, and the fork in the road

[Against the three criteria of Sec. 1.5, LD fails all three in one form or another. Two ways to modify it then present themselves, and they are the subjects of the next two sections:

- sideways — keep the structure of the reaction term and open up its coefficient (Sec. 4);
- upward — keep the coefficient and raise the order of the equation by enlarging B^μ (Sec. 5).

This contrast is the spine of the notes.]

4. SPLITTING OFF THE SINGULAR PART: THE λ - κ FAMILY

4.1. The equivalence theorem

[The motion obtained from momentum–energy conservation is the same as the motion obtained by applying the Lorentz force law with $\frac{1}{2}(A_{\text{ret}} - A_{\text{adv}})$. Give the elementary proof from the 2006 text (Taylor expansion of the retarded potential, term-by-term cancellation), which is more direct than the routes via Zhang Zongsui or Poisson.]

4.2. Does the singular part act?

[Construct the stress–energy tensor of $A_S = \frac{1}{2}(A_{\text{ret}} + A_{\text{adv}})$ and integrate it. Eq. (1.2.17) of the 2006 text:]

$$\frac{dP^\alpha}{d\tau} = \frac{a^\alpha}{2r} + \frac{2}{3}\dot{a}^\alpha. \quad (6)$$

[The 2006 text read the surviving second term as showing that the singular part is not inert, and used that as the licence to loosen the split. Flag: this is the load-bearing step of the whole section, and it needs to be checked — see the closing remark of Sec. 4.7.]

4.3. The two-parameter family

$$A^\alpha = \lambda A_{\text{ret}}^\alpha + \kappa A_{\text{adv}}^\alpha. \quad (7)$$

[Two branches are then singled out by requiring that the retained part have a sensible field-theoretic status.]

4.4. Branch (a): $\lambda + \kappa = 0$

[The effective part satisfies the homogeneous Maxwell equations. The free-particle a - v phase portrait has a closed-form solution:]

$$a = \left[-\frac{3}{8\kappa} \ln \frac{1+v}{1-v} + C \right] (1-v^2)^{3/2}. \quad (8)$$

[$\kappa = -1/2$ recovers LD. The resulting equation of motion is Eq. (1.5.1) of the 2006 text:]

$$ma^\alpha = F_{\text{ext}}^\alpha - \frac{4\kappa}{3}e^2 (\dot{a}^\alpha - a^2 u^\alpha). \quad (9)$$

[Behaviour: $\kappa < 0$ gives a convex curve and still runs away; $\kappa > 0$ gives a concave curve that must cross the v axis, so a perturbed particle decelerates back to uniform motion. The 2006 text called this “Aristotle motion,” since force then fixes velocity rather than acceleration.]

4.5. Branch (b): $1 - \lambda = -\kappa$

[Retarded and advanced parts enter the discarded singular term on equal footing. Solve numerically in κ . Result: $\kappa = -1$ is the watershed. For $\kappa > -1$ the runaway survives; for $\kappa < -1$ a critical velocity v_c appears — a particle starting below v_c stays below it, one starting above decelerates stably. Reproduce the table:]

κ	-1.5	-2	-3	-500	-5000
v_c	0.3473	0.2261	0.1341	6.67×10^{-3}	6.67×10^{-4}

4.6. An unresolved point

[Extending branch (b) beyond one dimension runs into $\mathbf{n} \cdot \mathbf{v}$ being undefined at the position of the charge itself. The 2006 text suggested the replacement $\mathbf{n} \cdot \mathbf{v} \rightarrow \frac{1}{3}v$ but did not carry it out.]

4.7. What this branch amounts to

[Eq. (9) is nothing but the LD equation with the radiation-reaction coefficient $2e^2/3$ replaced by a free parameter $-4\kappa e^2/3$. State this plainly — it is the one result in these notes that outlives its original motivation, whatever becomes of Eq. (6).]

5. THE ELIEZER-TYPE SERIES

5.1. A note on the name

[Placed first, not in a footnote. The equations of this section follow an ansatz for B^μ taken from Eliezer’s 1947 review. They are not the equation now standardly called the Eliezer, or Eliezer–Ford–O’Connell, equation, which is obtained by order reduction, is free of runaways, and is a live candidate in present-day experiments. Confusing the two inverts the conclusion of this section.]

5.2. Expanding B^μ

$$B_\mu = P_0 v_\mu + P_1 \dot{v}_\mu + P_2 \ddot{v}_\mu + \cdots + P_i v_\mu^{(i)} + \cdots \quad (10)$$

[Differentiating $v^\mu v_\mu = -1$ repeatedly generates a chain of identities whose coefficients form Pascal’s triangle; the general term is in Appendix A.4 of the 2006 text. Substituting gives the generalized equation of motion, Eq. (1.3.3) there. Setting all $P_i = 0$ returns LD, Eq. (3).]

5.3. Keeping P_2 : a third-order equation

$$\ddot{q} - B\ddot{q} + A\dot{q} - \frac{1}{2}\dot{q}^3 = 0, \quad A = \frac{3m}{2k}, \quad B = \frac{e^2}{k}. \quad (11)$$

[Scan the (A, B) plane numerically. Reported outcome: for A and B of either common sign the particle accelerates to c on its own; for $A, B > 0$ with $A \gg B$ the behaviour is worse still, the velocity swinging at high frequency between $+c$ and $-c$. Figures 1-3-1 through 1-3-4 of the 2006 text.]

5.4. Keeping P_2 and P_4 : a fifth-order equation

[Eq. (1.3.11) of the 2006 text, scanned over 22 parameter triples (A, B, C) . Three classes of behaviour: oscillation between $\pm c$; monotonic approach to c ; and a third class, neither reaching c nor returning to uniform motion, in which the speed pulses periodically within a bounded range — “much like an oscillator, except that the oscillator itself keeps moving forward.”]

5.5. Scorecard

[No parameter choice in either the third- or the fifth-order family satisfies the criteria of Sec. 1.5. Close with the question the 2006 text closed on — whether nature would really select a more complicated B^μ — and leave it open here.]